```
pwd
```

```
'C:\\Users\\Admin'
```

```
'C:\\Users\\Admin\\Downloads'
```

```
'C:\\Users\\Admin\\Downloads'
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data=pd.read_csv(r'C:\\Users\\Admin\\Downloads\\zomato.csv')
```

```
data.shape
```

```
(51717, 17)
```

```
data.head()
```

```
                                                 url  \
0  https://www.zomato.com/bangalore/jalsa-banasha...
1  https://www.zomato.com/bangalore/spice-elephan...
2  https://www.zomato.com/SanchurroBangalore?cont...
3  https://www.zomato.com/bangalore/addhuri-udupi...
4  https://www.zomato.com/bangalore/grand-village...

                                             address
name  \
0  942, 21st Main Road, 2nd Stage, Banashankari, ...
Jalsa
1  2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...            Spice
Elephant
2  1112, Next to KIMS Medical College, 17th Cross...        San Churro
Cafe
3  1st Floor, Annakuteera, 3rd Stage, Banashankar...  Addhuri Udupi
Bhojana
4  10, 3rd Floor, Lakshmi Associates, Gandhi Baza...           Grand
Village

  online_order book_table   rate  votes
phone  \
0          Yes        Yes  4.1/5    775    080 42297555\r\n+91
9743772233
1          Yes         No  4.1/5    787                      080
41714161
2          Yes         No  3.8/5    918                     +91
9663487993
3           No         No  3.7/5     88                     +91
```

```
9620009302
4          No         No  3.8/5    166  +91 8026612447\r\n+91
9901210005

       location            rest_type  \
0  Banashankari        Casual Dining
1  Banashankari        Casual Dining
2  Banashankari  Cafe, Casual Dining
3  Banashankari          Quick Bites
4  Basavanagudi        Casual Dining

                                          dish_liked  \
0  Pasta, Lunch Buffet, Masala Papad, Paneer Laja...
1  Momos, Lunch Buffet, Chocolate Nirvana, Thai G...
2  Churros, Cannelloni, Minestrone Soup, Hot Choc...
3                                        Masala Dosa
4                                Panipuri, Gol Gappe

                         cuisines approx_cost(for two people)  \
0  North Indian, Mughlai, Chinese                         800
1     Chinese, North Indian, Thai                         800
2          Cafe, Mexican, Italian                         800
3      South Indian, North Indian                         300
4        North Indian, Rajasthani                         600

                                        reviews_list menu_item  \
0  [('Rated 4.0', 'RATED\n  A beautiful place to ...        []
1  [('Rated 4.0', 'RATED\n  Had been here for din...        []
2  [('Rated 3.0', "RATED\n  Ambience is not that ...        []
3  [('Rated 4.0', "RATED\n  Great food and proper...        []
4  [('Rated 4.0', 'RATED\n  Very good restaurant ...        []

  listed_in(type) listed_in(city)
0          Buffet    Banashankari
1          Buffet    Banashankari
2          Buffet    Banashankari
3          Buffet    Banashankari
4          Buffet    Banashankari
```

```
data.columns
```

```
Index(['url', 'address', 'name', 'online_order', 'book_table', 'rate',
'votes',
       'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
       'approx_cost(for two people)', 'reviews_list', 'menu_item',
       'listed_in(type)', 'listed_in(city)'],
      dtype='object')
```

```
data.tail()
```

```
                                                     url  \
51712  https://www.zomato.com/bangalore/best-brews-fo...
51713  https://www.zomato.com/bangalore/vinod-bar-and...
51714  https://www.zomato.com/bangalore/plunge-sherat...
51715  https://www.zomato.com/bangalore/chime-sherato...
51716  https://www.zomato.com/bangalore/the-nest-the-...

                                                 address  \
51712  Four Points by Sheraton Bengaluru, 43/3, White...
51713  Number 10, Garudachar Palya, Mahadevapura, Whi...
51714  Sheraton Grand Bengaluru Whitefield Hotel & Co...
51715  Sheraton Grand Bengaluru Whitefield Hotel & Co...
51716  ITPL Main Road, KIADB Export Promotion Industr...

                                                    name online_order
\
51712  Best Brews - Four Points by Sheraton Bengaluru...           No

51713                         Vinod Bar And Restaurant           No

51714  Plunge - Sheraton Grand Bengaluru Whitefield H...           No

51715  Chime - Sheraton Grand Bengaluru Whitefield Ho...           No

51716                     The Nest - The Den Bengaluru           No


      book_table    rate  votes           phone
location  \
51712         No  3.6 /5     27    080 40301477
Whitefield
51713         No     NaN      0  +91 8197675843
Whitefield
51714         No     NaN      0             NaN
Whitefield
51715        Yes  4.3 /5    236    080 49652769  ITPL Main Road,
Whitefield
51716         No  3.4 /5     13  +91 8071117272  ITPL Main Road,
Whitefield

                rest_type                     dish_liked  \
51712                 Bar                            NaN
51713                 Bar                            NaN
51714                 Bar                            NaN
51715                 Bar  Cocktails, Pizza, Buttermilk
51716  Bar, Casual Dining                            NaN

                                     cuisines approx_cost(for two
people)  \
51712                             Continental
1,500
```

```
51713                              Finger Food
600
51714                              Finger Food
2,000
51715                              Finger Food
2,500
51716  Finger Food, North Indian, Continental
1,500

                                            reviews_list menu_item  \
51712  [('Rated 5.0', "RATED\n  Food and service are ...        []
51713                                                 []        []
51714                                                 []        []
51715  [('Rated 4.0', 'RATED\n  Nice and friendly pla...        []
51716  [('Rated 5.0', 'RATED\n  Great ambience , look...        []

      listed_in(type) listed_in(city)
51712   Pubs and bars      Whitefield
51713   Pubs and bars      Whitefield
51714   Pubs and bars      Whitefield
51715   Pubs and bars      Whitefield
51716   Pubs and bars      Whitefield
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51717 entries, 0 to 51716
Data columns (total 17 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51717 non-null  object
 1   address                      51717 non-null  object
 2   name                         51717 non-null  object
 3   online_order                 51717 non-null  object
 4   book_table                   51717 non-null  object
 5   rate                         43942 non-null  object
 6   votes                        51717 non-null  int64
 7   phone                        50509 non-null  object
 8   location                     51696 non-null  object
 9   rest_type                    51490 non-null  object
 10  dish_liked                   23639 non-null  object
 11  cuisines                     51672 non-null  object
 12  approx_cost(for two people)  51371 non-null  object
 13  reviews_list                 51717 non-null  object
 14  menu_item                    51717 non-null  object
 15  listed_in(type)              51717 non-null  object
 16  listed_in(city)              51717 non-null  object
dtypes: int64(1), object(16)
memory usage: 6.7+ MB
```

```
51717-23639

28078

data.isnull().sum()

url                                0
address                            0
name                               0
online_order                       0
book_table                         0
rate                            7775
votes                              0
phone                           1208
location                          21
rest_type                        227
dish_liked                     28078
cuisines                          45
approx_cost(for two people)      346
reviews_list                       0
menu_item                          0
listed_in(type)                    0
listed_in(city)                    0
dtype: int64

data1=data.drop(['url','dish_liked','phone'],axis=1)
#data.shape

data1.head()

                                             address
name  \
0  942, 21st Main Road, 2nd Stage, Banashankari, ...
Jalsa
1  2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...          Spice
Elephant
2  1112, Next to KIMS Medical College, 17th Cross...         San Churro
Cafe
3  1st Floor, Annakuteera, 3rd Stage, Banashankar...  Addhuri Udupi
Bhojana
4  10, 3rd Floor, Lakshmi Associates, Gandhi Baza...           Grand
Village

  online_order book_table   rate  votes      location
rest_type  \
0          Yes        Yes  4.1/5    775  Banashankari        Casual
Dining
1          Yes         No  4.1/5    787  Banashankari        Casual
Dining
2          Yes         No  3.8/5    918  Banashankari  Cafe, Casual
Dining
```

```
3           No         No  3.7/5     88  Banashankari         Quick
Bites
4           No         No  3.8/5    166  Basavanagudi        Casual
Dining

                         cuisines approx_cost(for two people)  \
0  North Indian, Mughlai, Chinese                         800
1     Chinese, North Indian, Thai                         800
2          Cafe, Mexican, Italian                         800
3       South Indian, North Indian                        300
4         North Indian, Rajasthani                        600

                                        reviews_list menu_item  \
0  [('Rated 4.0', 'RATED\n  A beautiful place to ...        []
1  [('Rated 4.0', 'RATED\n  Had been here for din...        []
2  [('Rated 3.0', "RATED\n  Ambience is not that ...        []
3  [('Rated 4.0', "RATED\n  Great food and proper...        []
4  [('Rated 4.0', 'RATED\n  Very good restaurant ...        []

  listed_in(type) listed_in(city)
0          Buffet    Banashankari
1          Buffet    Banashankari
2          Buffet    Banashankari
3          Buffet    Banashankari
4          Buffet    Banashankari
```

```
data1.head(3)
```

```
                                             address             name
\
0  942, 21st Main Road, 2nd Stage, Banashankari, ...            Jalsa

1  2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...   Spice Elephant

2  1112, Next to KIMS Medical College, 17th Cross...  San Churro Cafe


  online_order book_table   rate  votes      location
rest_type  \
0          Yes        Yes  4.1/5    775  Banashankari        Casual
Dining
1          Yes         No  4.1/5    787  Banashankari        Casual
Dining
2          Yes         No  3.8/5    918  Banashankari  Cafe, Casual
Dining

                         cuisines approx_cost(for two people)  \
0  North Indian, Mughlai, Chinese                         800
1     Chinese, North Indian, Thai                         800
```

```
2          Cafe, Mexican, Italian                        800

                                       reviews_list menu_item  \
0  [('Rated 4.0', 'RATED\n  A beautiful place to ...        []
1  [('Rated 4.0', 'RATED\n  Had been here for din...        []
2  [('Rated 3.0', "RATED\n  Ambience is not that ...        []

  listed_in(type) listed_in(city)
0          Buffet    Banashankari
1          Buffet    Banashankari
2          Buffet    Banashankari
```

```
data1.duplicated().sum()
```

```
43
```

```
51717-43
```

```
51674
```

```
data1.drop_duplicates(inplace=True)
```

```
data1.shape
```

```
(51674, 14)
```

```
data1.isnull()
```

```
       address   name  online_order  book_table   rate  votes
location  \
0        False  False         False       False  False  False
False
1        False  False         False       False  False  False
False
2        False  False         False       False  False  False
False
3        False  False         False       False  False  False
False
4        False  False         False       False  False  False
False
...        ...    ...           ...         ...    ...    ...        ..
.
51712    False  False         False       False  False  False
False
51713    False  False         False       False   True  False
False
51714    False  False         False       False   True  False
False
51715    False  False         False       False  False  False
False
51716    False  False         False       False  False  False
```

```
False

       rest_type  cuisines  approx_cost(for two people)  reviews_list
\
0          False     False                        False         False

1          False     False                        False         False

2          False     False                        False         False

3          False     False                        False         False

4          False     False                        False         False

...          ...       ...                          ...           ...

51712      False     False                        False         False

51713      False     False                        False         False

51714      False     False                        False         False

51715      False     False                        False         False

51716      False     False                        False         False


       menu_item  listed_in(type)  listed_in(city)
0          False            False            False
1          False            False            False
2          False            False            False
3          False            False            False
4          False            False            False
...          ...              ...              ...
51712      False            False            False
51713      False            False            False
51714      False            False            False
51715      False            False            False
51716      False            False            False

[51674 rows x 14 columns]
```

```
data1.isnull().sum()
```

```
address                            0
name                               0
online_order                       0
book_table                         0
rate                            7767
votes                              0
location                          21
```

```
rest_type                        227
cuisines                          45
approx_cost(for two people)      345
reviews_list                       0
menu_item                          0
listed_in(type)                    0
listed_in(city)                    0
dtype: int64
```

```
7767+21+227+45+345
```

```
8405
```

```
51674-8405
```

```
43269
```

```
data1.dropna(how='any',inplace=True)
```

```
data1.shape
```

```
(43499, 14)
```

```
data1.isnull()
```

```
       address   name  online_order  book_table   rate  votes  
location  \
0        False  False         False       False  False  False  
False
1        False  False         False       False  False  False  
False
2        False  False         False       False  False  False  
False
3        False  False         False       False  False  False  
False
4        False  False         False       False  False  False  
False
...        ...    ...           ...         ...    ...    ...        ..
.
51709    False  False         False       False  False  False  
False
51711    False  False         False       False  False  False  
False
51712    False  False         False       False  False  False  
False
51715    False  False         False       False  False  False  
False
51716    False  False         False       False  False  False  
False

       rest_type  cuisines  approx_cost(for two people)  reviews_list
```

```
\
0          False     False                            False        False

1          False     False                            False        False

2          False     False                            False        False

3          False     False                            False        False

4          False     False                            False        False

...          ...       ...                              ...          ...

51709      False     False                            False        False

51711      False     False                            False        False

51712      False     False                            False        False

51715      False     False                            False        False

51716      False     False                            False        False


       menu_item  listed_in(type)  listed_in(city)
0          False            False            False
1          False            False            False
2          False            False            False
3          False            False            False
4          False            False            False
...          ...              ...              ...
51709      False            False            False
51711      False            False            False
51712      False            False            False
51715      False            False            False
51716      False            False            False

[43499 rows x 14 columns]
```

```
data1.isnull().sum()
```

```
address                        0
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
location                       0
rest_type                      0
cuisines                       0
approx_cost(for two people)    0
```

```
reviews_list                     0
menu_item                        0
listed_in(type)                  0
listed_in(city)                  0
dtype: int64
```

```
data1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 43499 entries, 0 to 51716
Data columns (total 14 columns):
 #   Column                       Non-Null Count  Dtype 
---  ------                       --------------  ----- 
 0   address                      43499 non-null  object
 1   name                         43499 non-null  object
 2   online_order                 43499 non-null  object
 3   book_table                   43499 non-null  object
 4   rate                         43499 non-null  object
 5   votes                        43499 non-null  int64 
 6   location                     43499 non-null  object
 7   rest_type                    43499 non-null  object
 8   cuisines                     43499 non-null  object
 9   approx_cost(for two people)  43499 non-null  object
 10  reviews_list                 43499 non-null  object
 11  menu_item                    43499 non-null  object
 12  listed_in(type)              43499 non-null  object
 13  listed_in(city)              43499 non-null  object
dtypes: int64(1), object(13)
memory usage: 5.0+ MB
```

```
data1.columns
```

```
Index(['address', 'name', 'online_order', 'book_table', 'rate', 
'votes',
       'location', 'rest_type', 'cuisines', 'approx_cost(for two 
people)',
       'reviews_list', 'menu_item', 'listed_in(type)', 
'listed_in(city)'],
      dtype='object')
```

```
data2=data1.rename(columns={'approx_cost(for two 
people)':'cost','listed_in(type)':'type','listed_in(city)':'city'})
```

```
data2.columns
```

```
Index(['address', 'name', 'online_order', 'book_table', 'rate', 
'votes',
       'location', 'rest_type', 'cuisines', 'cost', 'reviews_list',
       'menu_item', 'type', 'city'],
      dtype='object')
```

```
data2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 43499 entries, 0 to 51716
Data columns (total 14 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   address          43499 non-null  object
 1   name             43499 non-null  object
 2   online_order     43499 non-null  object
 3   book_table       43499 non-null  object
 4   rate             43499 non-null  object
 5   votes            43499 non-null  int64
 6   location         43499 non-null  object
 7   rest_type        43499 non-null  object
 8   cuisines         43499 non-null  object
 9   cost             43499 non-null  object
 10  reviews_list     43499 non-null  object
 11  menu_item        43499 non-null  object
 12  listed_in(type)  43499 non-null  object
 13  city             43499 non-null  object
dtypes: int64(1), object(13)
memory usage: 6.0+ MB
```

```
data2['cost']
```

```
0          800
1          800
2          800
3          300
4          600
         ...
51709      800
51711      800
51712    1,500
51715    2,500
51716    1,500
Name: cost, Length: 43499, dtype: object
```

```
data2['cost']=data2['cost'].astype(str)
data2['cost']=data2['cost'].apply(lambda x:x.replace(',','.'))
```

```
data2['cost']
```

```
0          800
1          800
2          800
3          300
4          600
         ...
51709      800
```

```
51711      800
51712    1.500
51715    2.500
51716    1.500
Name: cost, Length: 43499, dtype: object
```

```python
data2['cost']=data2['cost'].astype(float)
```

```python
data2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 43499 entries, 0 to 51716
Data columns (total 14 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   address        43499 non-null  object
 1   name           43499 non-null  object
 2   online_order   43499 non-null  object
 3   book_table     43499 non-null  object
 4   rate           43499 non-null  object
 5   votes          43499 non-null  int64
 6   location       43499 non-null  object
 7   rest_type      43499 non-null  object
 8   cuisines       43499 non-null  object
 9   cost           43499 non-null  float64
 10  reviews_list   43499 non-null  object
 11  menu_item      43499 non-null  object
 12  type           43499 non-null  object
 13  city           43499 non-null  object
dtypes: float64(1), int64(1), object(12)
memory usage: 6.0+ MB
```

```python
data2['rate'].unique()
```

```
array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
       '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
       '4.3/5', 'NEW', '2.9/5', '3.5/5', '2.6/5', '3.8 /5', '3.4/5',
       '4.5/5', '2.5/5', '2.7/5', '4.7/5', '2.4/5', '2.2/5', '2.3/5',
       '3.4 /5', '-', '3.6 /5', '4.8/5', '3.9 /5', '4.2 /5', '4.0 /5',
       '4.1 /5', '3.7 /5', '3.1 /5', '2.9 /5', '3.3 /5', '2.8 /5',
       '3.5 /5', '2.7 /5', '2.5 /5', '3.2 /5', '2.6 /5', '4.5 /5',
       '4.3 /5', '4.4 /5', '4.9/5', '2.1/5', '2.0/5', '1.8/5', '4.6
/5',
       '4.9 /5', '3.0 /5', '4.8 /5', '2.3 /5', '4.7 /5', '2.4 /5',
       '2.1 /5', '2.2 /5', '2.0 /5', '1.8 /5'], dtype=object)
```

```python
data3=data2[data2['rate']!='NEW']
```

```python
data3.shape
```

```
(41302, 14)
```

```
data4=data3[data3['rate']!='-']

data4.shape

(41237, 14)

data4['rate'].unique()

array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
       '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
       '4.3/5', '2.9/5', '3.5/5', '2.6/5', '3.8 /5', '3.4/5', '4.5/5',
       '2.5/5', '2.7/5', '4.7/5', '2.4/5', '2.2/5', '2.3/5', '3.4 /5',
       '3.6 /5', '4.8/5', '3.9 /5', '4.2 /5', '4.0 /5', '4.1 /5',
       '3.7 /5', '3.1 /5', '2.9 /5', '3.3 /5', '2.8 /5', '3.5 /5',
       '2.7 /5', '2.5 /5', '3.2 /5', '2.6 /5', '4.5 /5', '4.3 /5',
       '4.4 /5', '4.9/5', '2.1/5', '2.0/5', '1.8/5', '4.6 /5', '4.9
/5',
       '3.0 /5', '4.8 /5', '2.3 /5', '4.7 /5', '2.4 /5', '2.1 /5',
       '2.2 /5', '2.0 /5', '1.8 /5'], dtype=object)

lambda x:x.replace('/5','') if type(x)==str else x

<function __main__.<lambda>(x)>

remove_slash=lambda x:x.replace('/s','') if type(x)==str else x



data4['rate']=data4['rate'].astype(str)
data4['rate']=data4['rate'].str.replace('/5','')
data4['rate']=data4['rate'].str.replace('','')
data['rate'] = pd.to_numeric(data['rate'],errors='coerce')
data['rate'].head()

C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\212150456.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4['rate']=data4['rate'].astype(str)
C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\212150456.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4['rate']=data4['rate'].str.replace('/5','')
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\212150456.py:3:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4['rate']=data4['rate'].str.replace('','')

0   NaN
1   NaN
2   NaN
3   NaN
4   NaN
Name: rate, dtype: float64
```

```
data4.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 41237 entries, 0 to 51716
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   address       41237 non-null  object
 1   name          41237 non-null  object
 2   online_order  41237 non-null  object
 3   book_table    41237 non-null  object
 4   rate          41237 non-null  object
 5   votes         41237 non-null  int64
 6   location      41237 non-null  object
 7   rest_type     41237 non-null  object
 8   cuisines      41237 non-null  object
 9   cost          41237 non-null  float64
 10  reviews_list  41237 non-null  object
 11  menu_item     41237 non-null  object
 12  type          41237 non-null  object
 13  city          41237 non-null  object
dtypes: float64(1), int64(1), object(12)
memory usage: 4.7+ MB
```

```
data['rate'].unique()
data.dtypes
```

```
url                                 object
address                             object
name                                object
online_order                        object
book_table                          object
rate                               float64
```

```
votes                                int64
phone                               object
location                            object
rest_type                           object
dish_liked                          object
cuisines                            object
approx_cost(for two people)         object
reviews_list                        object
menu_item                           object
listed_in(type)                     object
listed_in(city)                     object
dtype: object
```

```
data2['cost']=data2['cost'].astype(str)
data2['cost']=data2['cost'].apply(lambda x:x.replace(',','.'))
data2['cost']=data2['cost'].astype(float)
```

```
data4['rate']
```

```
0         4.1
1         4.1
2         3.8
3         3.7
4         3.8
         ...
51709    3.7
51711    2.5
51712    3.6
51715    4.3
51716    3.4
Name: rate, Length: 41237, dtype: object
```

```
data4.head()
```

```
                                             address
name  \
0  942, 21st Main Road, 2nd Stage, Banashankari, ...
Jalsa
1  2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...         Spice
Elephant
2  1112, Next to KIMS Medical College, 17th Cross...        San Churro
Cafe
3  1st Floor, Annakuteera, 3rd Stage, Banashankar...  Addhuri Udupi
Bhojana
4  10, 3rd Floor, Lakshmi Associates, Gandhi Baza...          Grand
Village

  online_order book_table rate  votes      location
rest_type  \
0          Yes        Yes  4.1    775  Banashankari        Casual
```

```
Dining
1          Yes          No  4.1    787  Banashankari        Casual
Dining
2          Yes          No  3.8    918  Banashankari  Cafe, Casual
Dining
3           No          No  3.7     88  Banashankari          Quick
Bites
4           No          No  3.8    166  Basavanagudi        Casual
Dining

                        cuisines   cost  \
0  North Indian, Mughlai, Chinese  800.0
1     Chinese, North Indian, Thai  800.0
2          Cafe, Mexican, Italian  800.0
3       South Indian, North Indian  300.0
4        North Indian, Rajasthani  600.0

                                        reviews_list menu_item    type
\
0  [('Rated 4.0', 'RATED\n  A beautiful place to ...        []  Buffet

1  [('Rated 4.0', 'RATED\n  Had been here for din...        []  Buffet

2  [('Rated 3.0', "RATED\n  Ambience is not that ...        []  Buffet

3  [('Rated 4.0', "RATED\n  Great food and proper...        []  Buffet

4  [('Rated 4.0', 'RATED\n  Very good restaurant ...        []  Buffet


           city
0  Banashankari
1  Banashankari
2  Banashankari
3  Banashankari
4  Banashankari
```

```
data4['name']=data4['name'].apply(lambda x:x.title())
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\3869025873.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4['name']=data4['name'].apply(lambda x:x.title())
```

```
data4['online_order'] .replace(('yes','No'),(True,False),inplace=True)
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\4158043554.py:1:
FutureWarning: A value is trying to be set on a copy of a DataFrame or
Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try
using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the
original object.


  data4['online_order'] .replace(('yes','No'),
(True,False),inplace=True)
C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\4158043554.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4['online_order'] .replace(('yes','No'),
(True,False),inplace=True)

data4['name'].factorize()

(array([   0,    1,    2, ..., 6532, 6568, 6569], dtype=int64),
 Index(['Jalsa', 'Spice Elephant', 'San Churro Cafe', 'Addhuri Udupi
Bhojana',
        'Grand Village', 'Timepass Dinner',
        'Rosewood International Hotel - Bar & Restaurant', 'Onesta',
        'Penthouse Cafe', 'Smacznego',
        ...
        'Food Dig', 'Ishwaryam', 'Hungry Wok', 'Mad Kitchen', 'Polly'S
Kitchen',
        'Calcutta North Indian Meals',
        'Chime - Sheraton Grand Bengaluru Whitefield Hotel &...',
        'The Nest - The Den Bengaluru', 'Nawabs Empire', 'Seeya
Restaurant'],
       dtype='object', length=6572))

for column in
data4.columns[data4.columns.isin(['rate','cost','votes'])]:
  print(column)


rate
votes
cost
```

```
for columns in
data4.columns[data4.columns.isin(['rate','cost','votes'])]:
    data4[columns]=data4[column].factorize()[0]


C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\22069510.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4[columns]=data4[column].factorize()[0]

data4['name'].factorize()[0]

import pandas as pd

def encode_all(df):
    df = df.copy()
    for col in df.columns:
        df[col] = pd.factorize(df[col])[0]
    return df

encoded_data4 = encode_all(data4)
encoded_data4.head()

   address  name  online_order  book_table  rate  votes  location
rest_type  \
0        0     0             0           0     0      0         0
0
1        1     1             0           1     0      0         0
0
2        2     2             0           1     0      0         0
1
3        3     3             1           1     1      1         0
2
4        4     4             1           1     2      2         1
0

   cuisines  cost  reviews_list  menu_item  type  city
0         0     0             0          0     0     0
1         1     0             1          0     0     0
2         2     0             2          0     0     0
3         3     1             3          0     0     0
4         4     2             4          0     0     0

encoded_data4.dtypes
```

```
address         int64
name            int64
online_order    int64
book_table      int64
rate            int64
votes           int64
location        int64
rest_type       int64
cuisines        int64
cost            int64
reviews_list    int64
menu_item       int64
type            int64
city            int64
dtype: object
```

```
data4_copy=data4.copy()
```

```
encoded_data4=encode(data4)
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_17176\2200031121.py:3:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  data4[columns]=data4[column].factorize()[0]
```

```
encoded_data4
```

```
       address  name  online_order  book_table  rate  votes  location
\
0            0     0             0           0     0      0         0

1            1     1             0           1     0      0         0

2            2     2             0           1     0      0         0

3            3     3             1           1     1      1         0

4            4     4             1           1     2      2         1

...        ...   ...           ...         ...   ...    ...       ...

51709     3137  2699             1           1     0      0        25

51711     8791  1716             1           1     0      0        25

51712     8725  6532             1           1    20     20        25
```

```
51715     8786  6568             1           0    36     36         56
51716     3444  6569             1           1    20     20         56

       rest_type  cuisines  cost  reviews_list  menu_item  type  city
0              0         0     0             0          0     0     0
1              0         1     0             1          0     0     0
2              1         2     0             2          0     0     0
3              2         3     1             3          0     0     0
4              0         4     2             4          0     0     0
...          ...       ...   ...           ...        ...   ...   ...
51709         28       204     0          4028          0     6    29
51711         28       761     0         21082          0     6    29
51712         17       240    20         20956          0     6    29
51715         17       237    36         21054          0     6    29
51716         33      1870    20         21055          0     6    29

[41237 rows x 14 columns]
```

```
encoded_data4.corr()
```

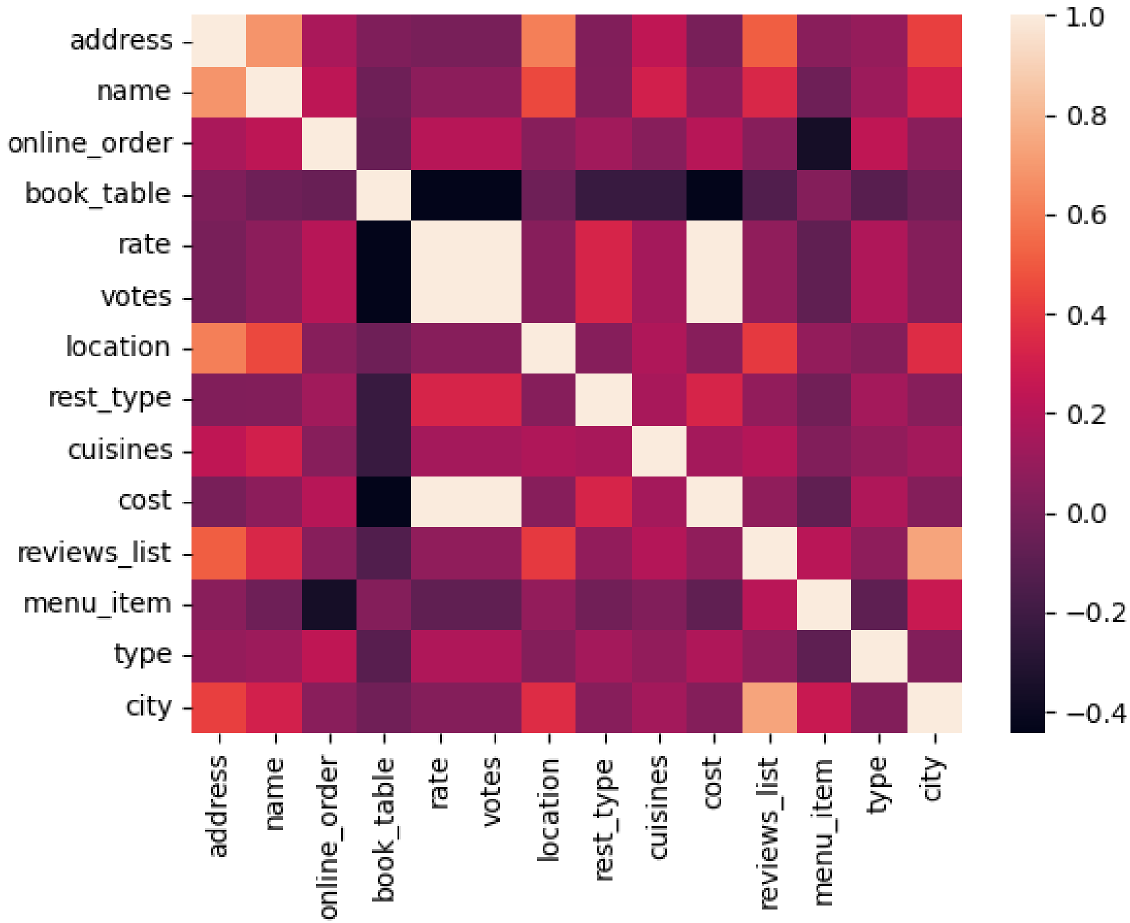
```
               address      name  online_order  book_table
rate  \
address       1.000000  0.682346      0.165707    0.021447 -0.002497
name          0.682346  1.000000      0.231616   -0.033718  0.070079
online_order  0.165707  0.231616      1.000000   -0.054894  0.205508
book_table    0.021447 -0.033718     -0.054894    1.000000 -0.442502
rate         -0.002497  0.070079      0.205508   -0.442502  1.000000
votes        -0.002497  0.070079      0.205508   -0.442502  1.000000
location      0.614626  0.448705      0.048645   -0.032781  0.051817
rest_type     0.030634  0.036212      0.136741   -0.226763  0.323978
```

```
cuisines       0.238421  0.301793      0.049921   -0.225396  0.148911

cost          -0.002497  0.070079      0.205508   -0.442502  1.000000

reviews_list   0.512074  0.339130      0.051440   -0.132809  0.078821

menu_item      0.058929 -0.031798     -0.362114    0.040496 -0.085893

type           0.094137  0.116000      0.239236   -0.114302  0.181699

city           0.423011  0.307128      0.054354   -0.028847  0.040517


                  votes  location  rest_type  cuisines      cost
reviews_list  \
address       -0.002497  0.614626   0.030634  0.238421 -0.002497
0.512074
name           0.070079  0.448705   0.036212  0.301793  0.070079
0.339130
online_order   0.205508  0.048645   0.136741  0.049921  0.205508
0.051440
book_table    -0.442502 -0.032781  -0.226763 -0.225396 -0.442502      -
0.132809
rate           1.000000  0.051817   0.323978  0.148911  1.000000
0.078821
votes          1.000000  0.051817   0.323978  0.148911  1.000000
0.078821
location       0.051817  1.000000   0.046133  0.178518  0.051817
0.407854
rest_type      0.323978  0.046133   1.000000  0.158553  0.323978
0.085501
cuisines       0.148911  0.178518   0.158553  1.000000  0.148911
0.196259
cost           1.000000  0.051817   0.323978  0.148911  1.000000
0.078821
reviews_list   0.078821  0.407854   0.085501  0.196259  0.078821
1.000000
menu_item     -0.085893  0.089058  -0.023399  0.026915 -0.085893
0.214814
type           0.181699  0.040451   0.147835  0.083311  0.181699
0.073885
city           0.040517  0.361957   0.047756  0.141997  0.040517
0.738147

              menu_item      type      city
address        0.058929  0.094137  0.423011
name          -0.031798  0.116000  0.307128
online_order  -0.362114  0.239236  0.054354
book_table     0.040496 -0.114302 -0.028847
```

```
rate            -0.085893  0.181699  0.040517
votes           -0.085893  0.181699  0.040517
location         0.089058  0.040451  0.361957
rest_type       -0.023399  0.147835  0.047756
cuisines         0.026915  0.083311  0.141997
cost            -0.085893  0.181699  0.040517
reviews_list     0.214814  0.073885  0.738147
menu_item        1.000000 -0.091713  0.271634
type            -0.091713  1.000000  0.034016
city             0.271634  0.034016  1.000000
```

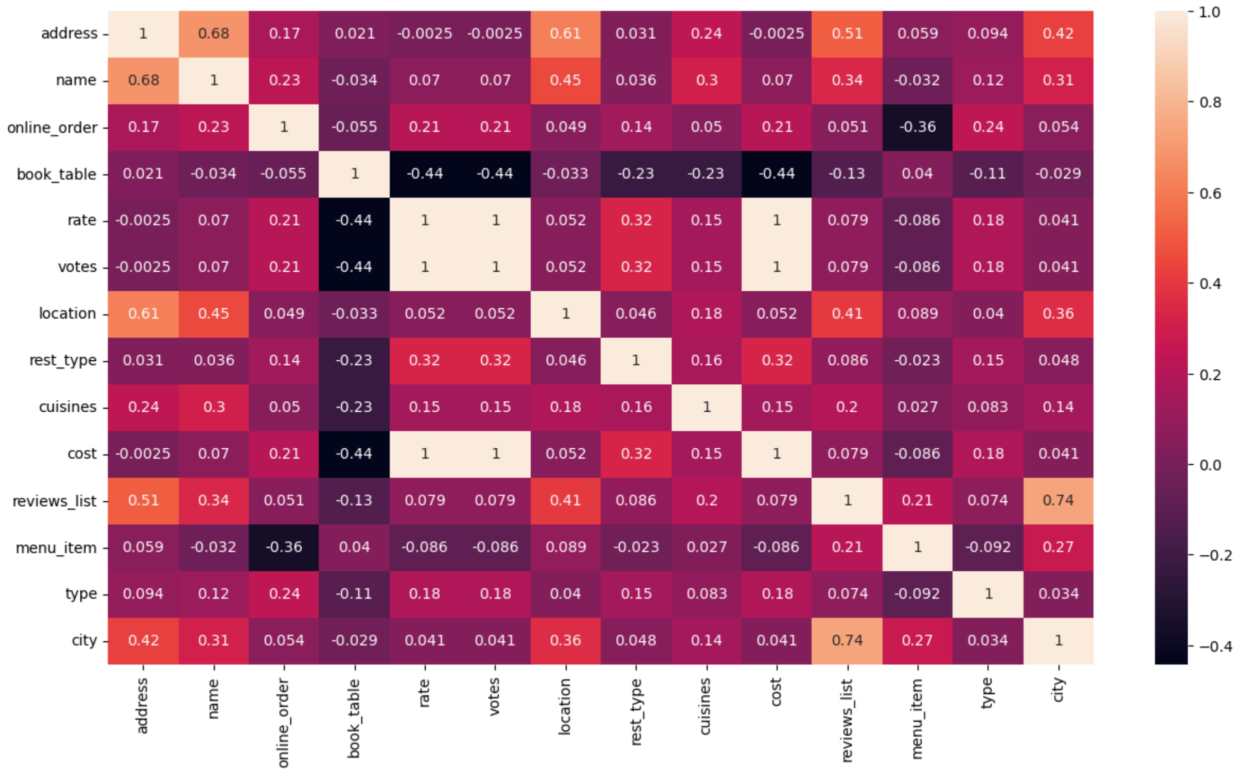
```
sns.heatmap(encoded_data4.corr())
```

```
<Axes: >
```



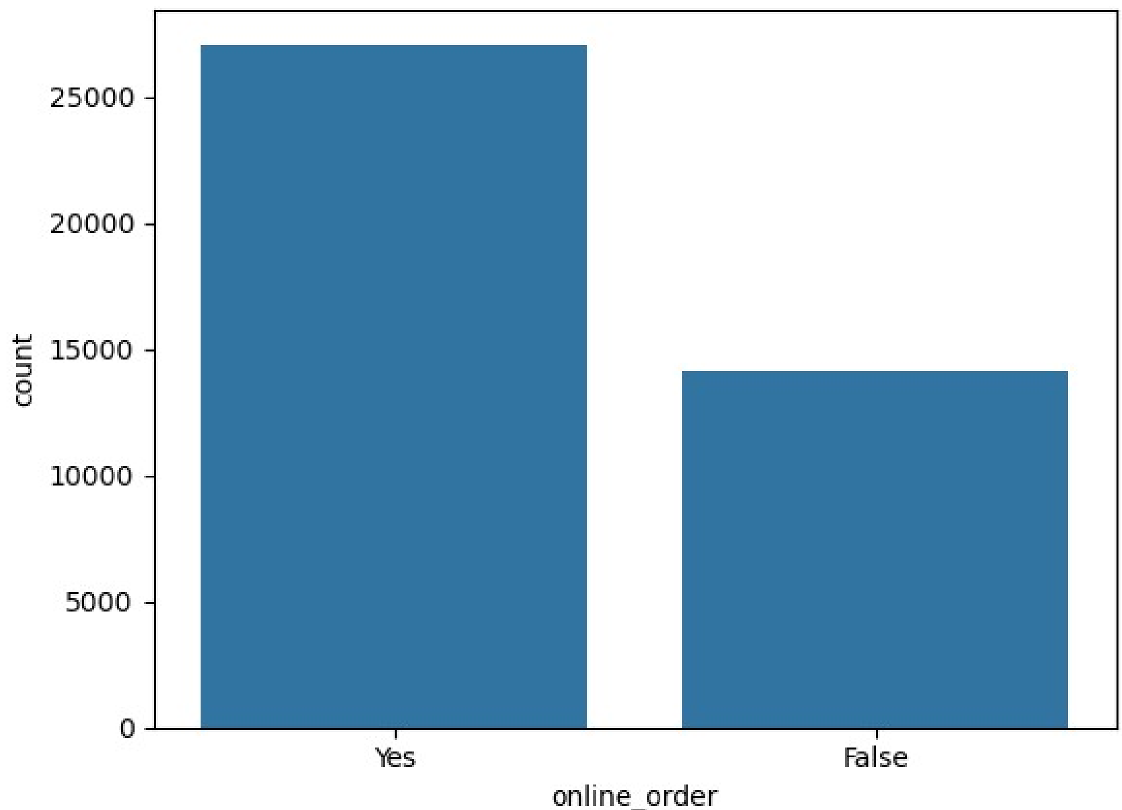

```
plt.figure(figsize=(15,8))
sns.heatmap(encoded_data4.corr(),annot=True)
```

```
<Axes: >
```




```
sns.countplot(x=data4_copy['online_order'])
```

```
<Axes: xlabel='online_order', ylabel='count'>
```



25000
20000
15000
10000
5000
0
count
Yes
False
online_order